



Sinh hoạt Khoa học hàng tuần tại bệnh viện Thống Nhất



Lecture 1

Boxplot and its Variants

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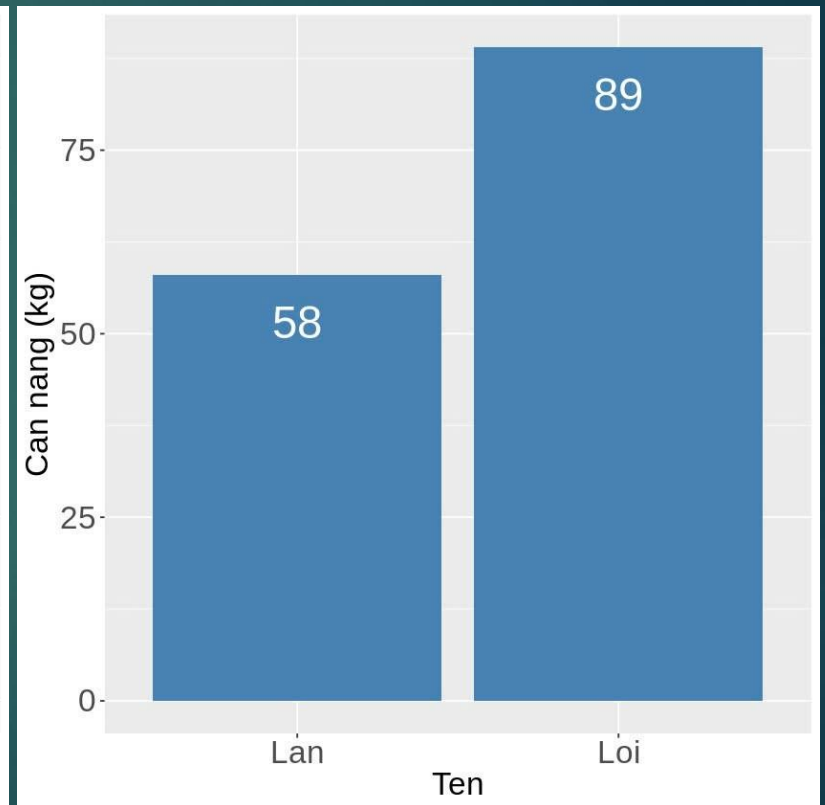
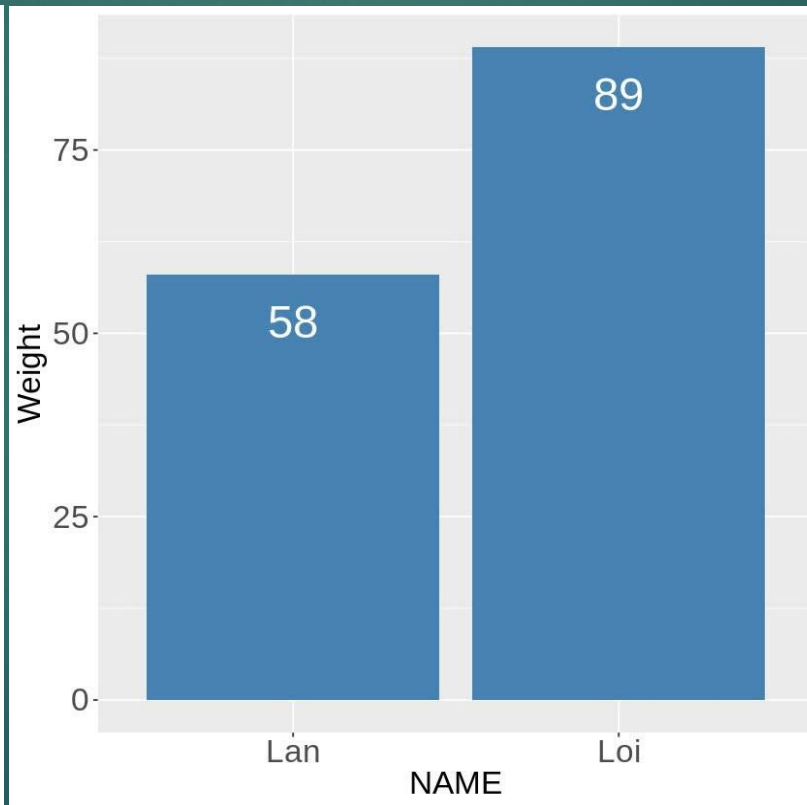
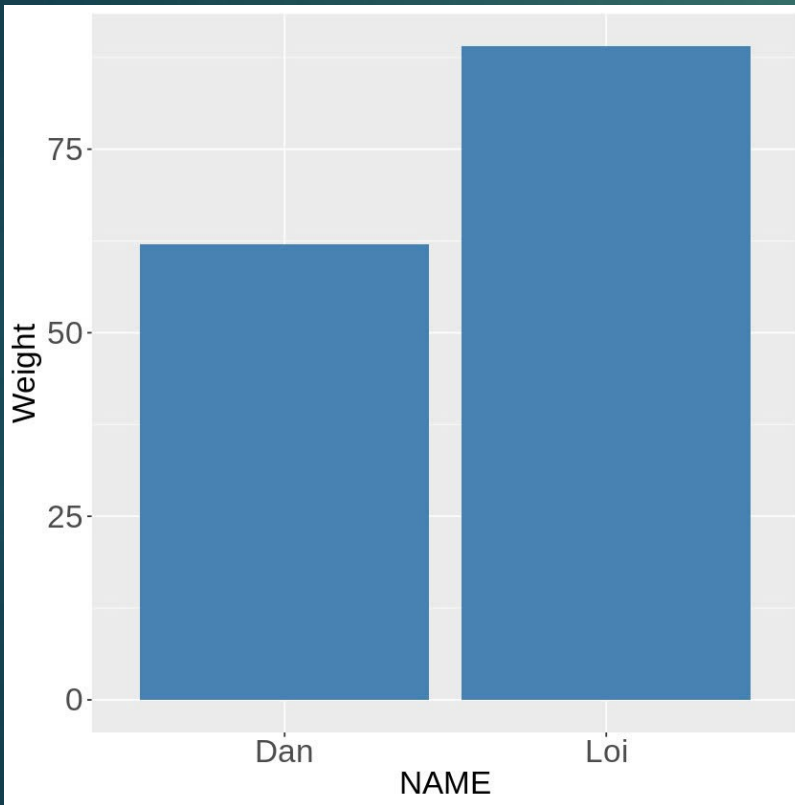
VIỆN NGHIÊN CỨU ỨNG DỤNG KHOA HỌC SỨC KHỎE VÀ LÃO HÓA – BỆNH VIỆN THỐNG NHẤT

CONTENT

- ▶ Barplot
- ▶ Boxplot: how to read
- ▶ Boxplot and its variants
- ▶ Violinplot
- ▶ Beanplot

Bar Plot

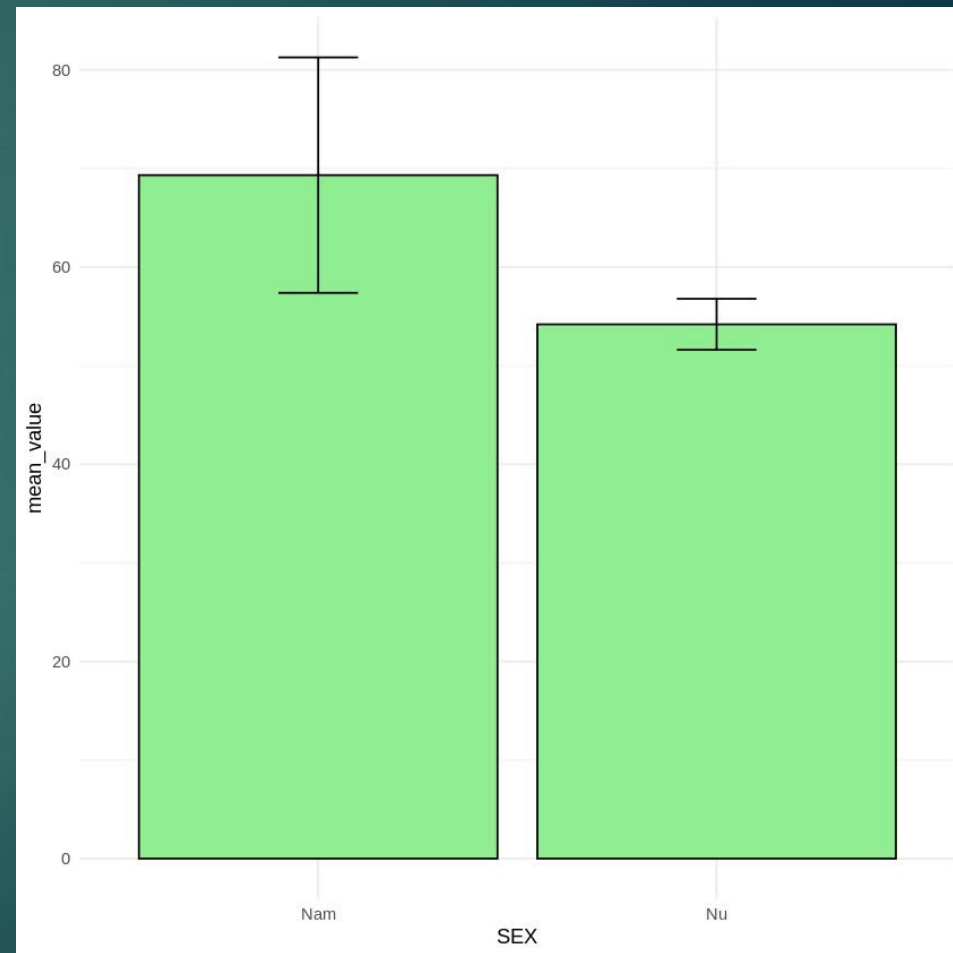
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1	Lan	Nu	24	1.63	58
2	Loi	Nam	43	1.73	89



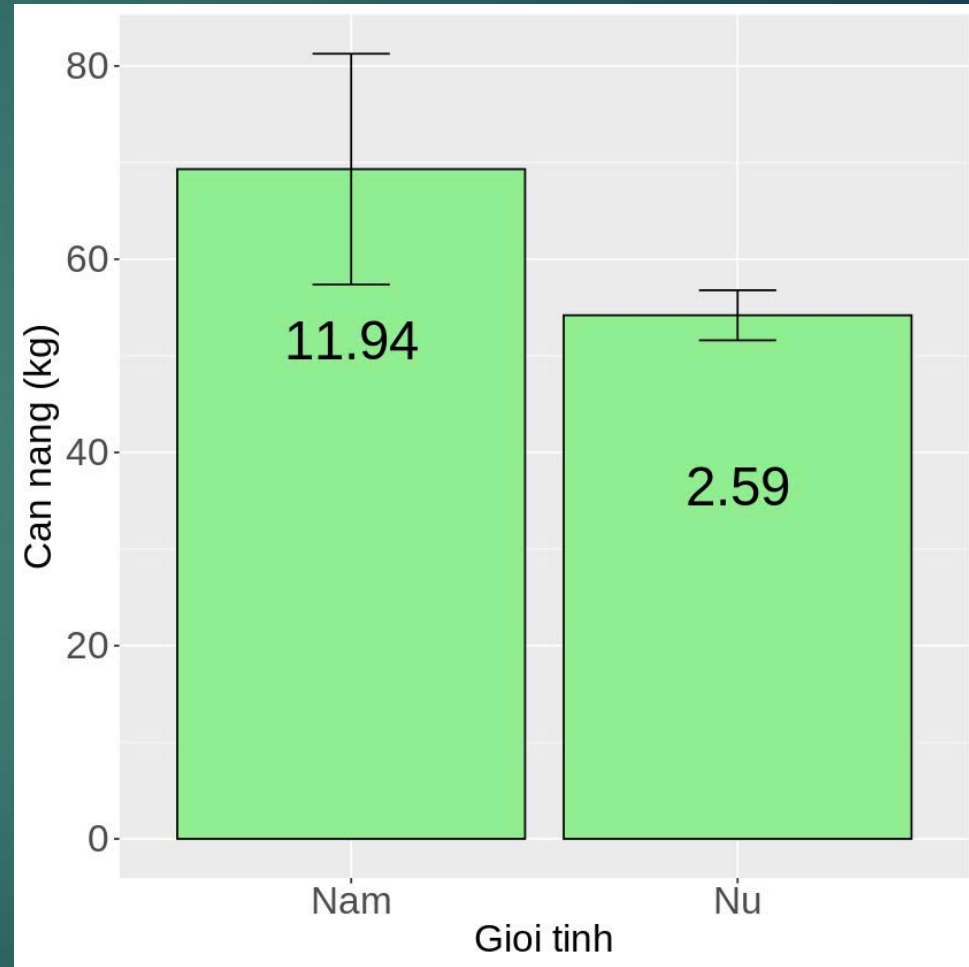
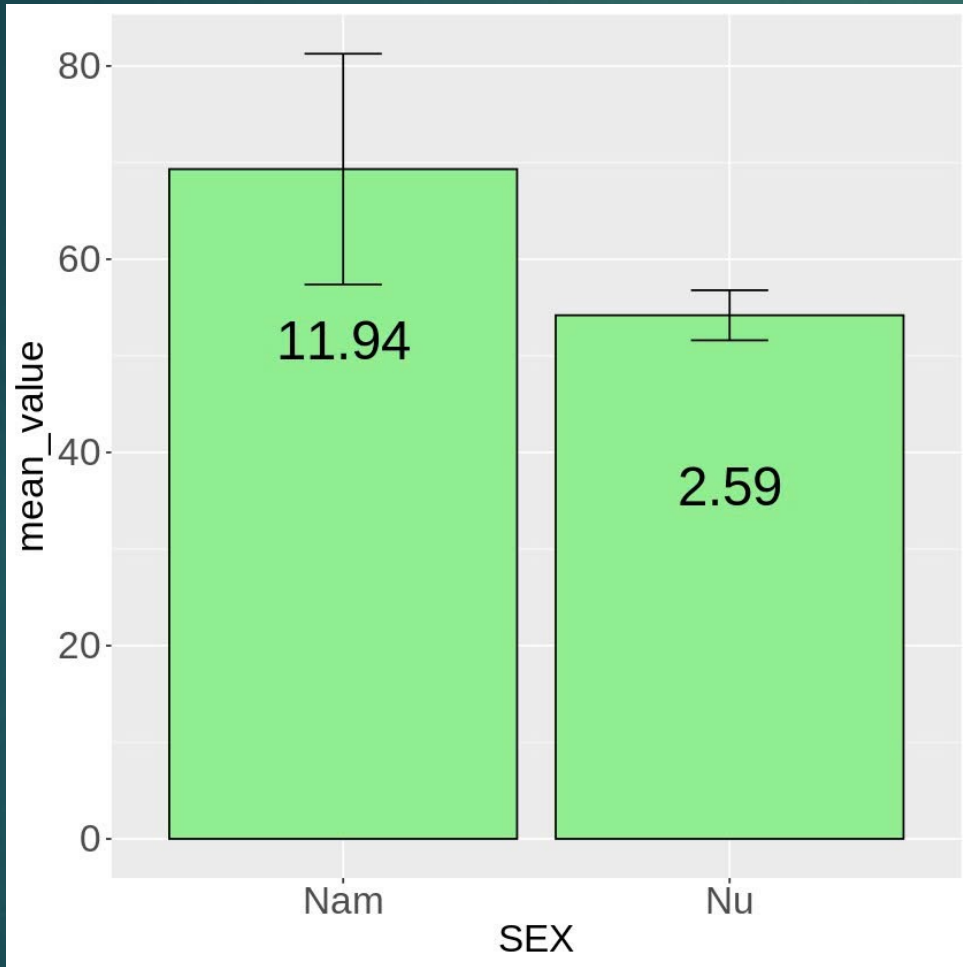
Bar Plot with Error Bar



No	NAME	SEX	AGE	Height	Weight
1	Dan	Nam	24	1.75	62
2	Loi	Nam	43	1.73	89
3	Uyen	Nu	25	1.62	54
4	Anh	Nu	45	1.56	53
5	Ngoc	Nu	40	1.58	55
6	Tai	Nam	48	1.68	67
7	Hoai	Nu	41	1.61	58
8	Hoang	Nam	23	1.70	69
9	Tung	Nam	45	1.78	75
10	Phi	Nam	23	1.69	54
11	Hai	Nu	36	1.66	51



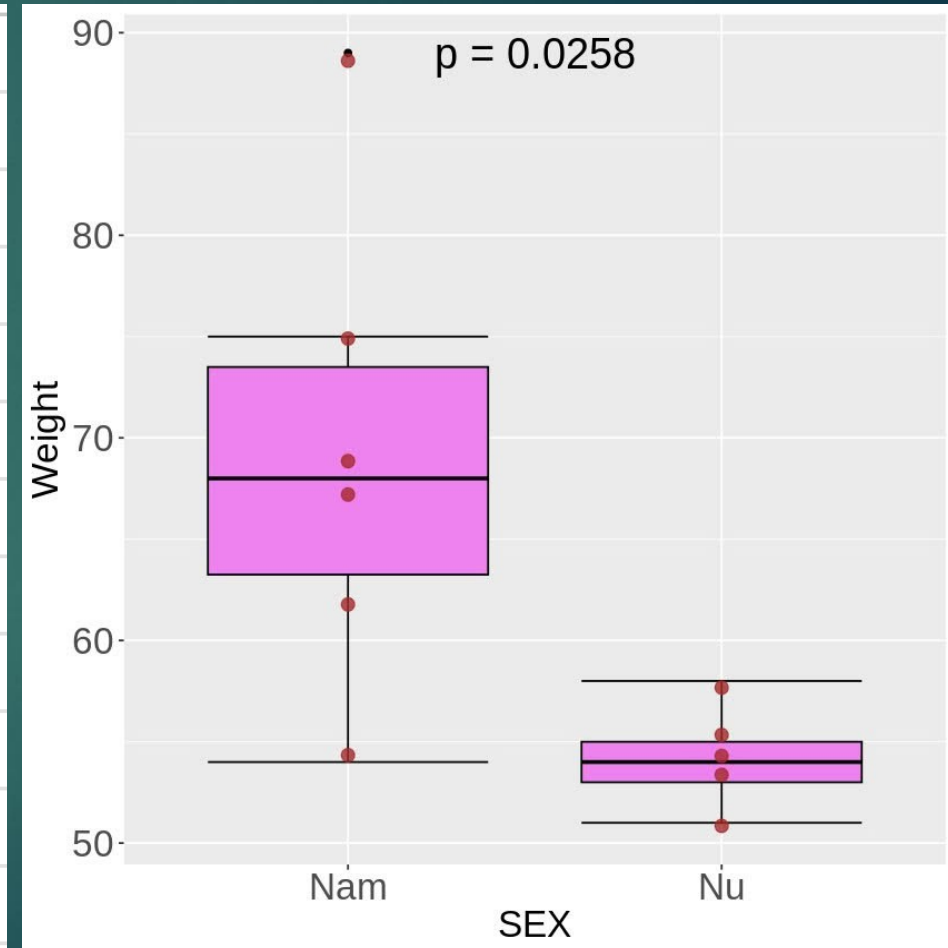
Bar Plot with Error Bar



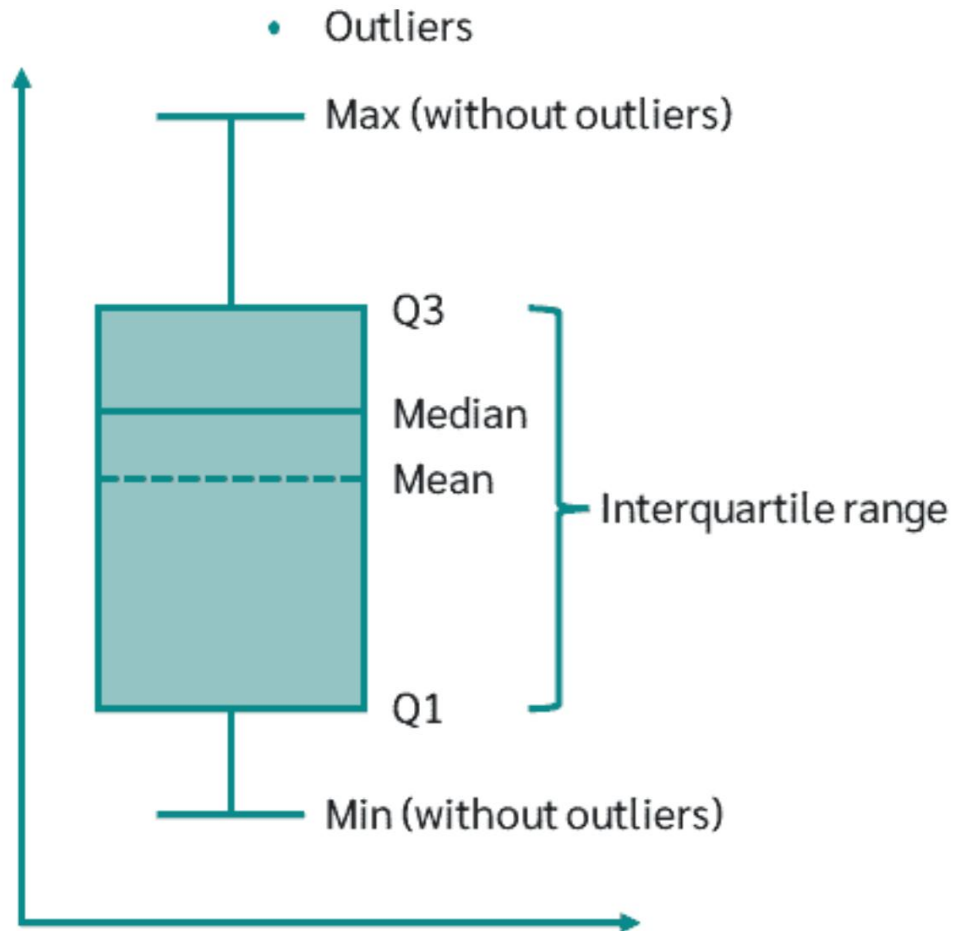
Boxplot



No	NAME	SEX	AGE	Height	Weight
1	Dan	Nam	24	1.75	62
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Boxplot



The box indicates the range in which the middle 50% of all data lies

Thus, the lower end of the box is the 1st quartile and the upper end is the 3rd quartile

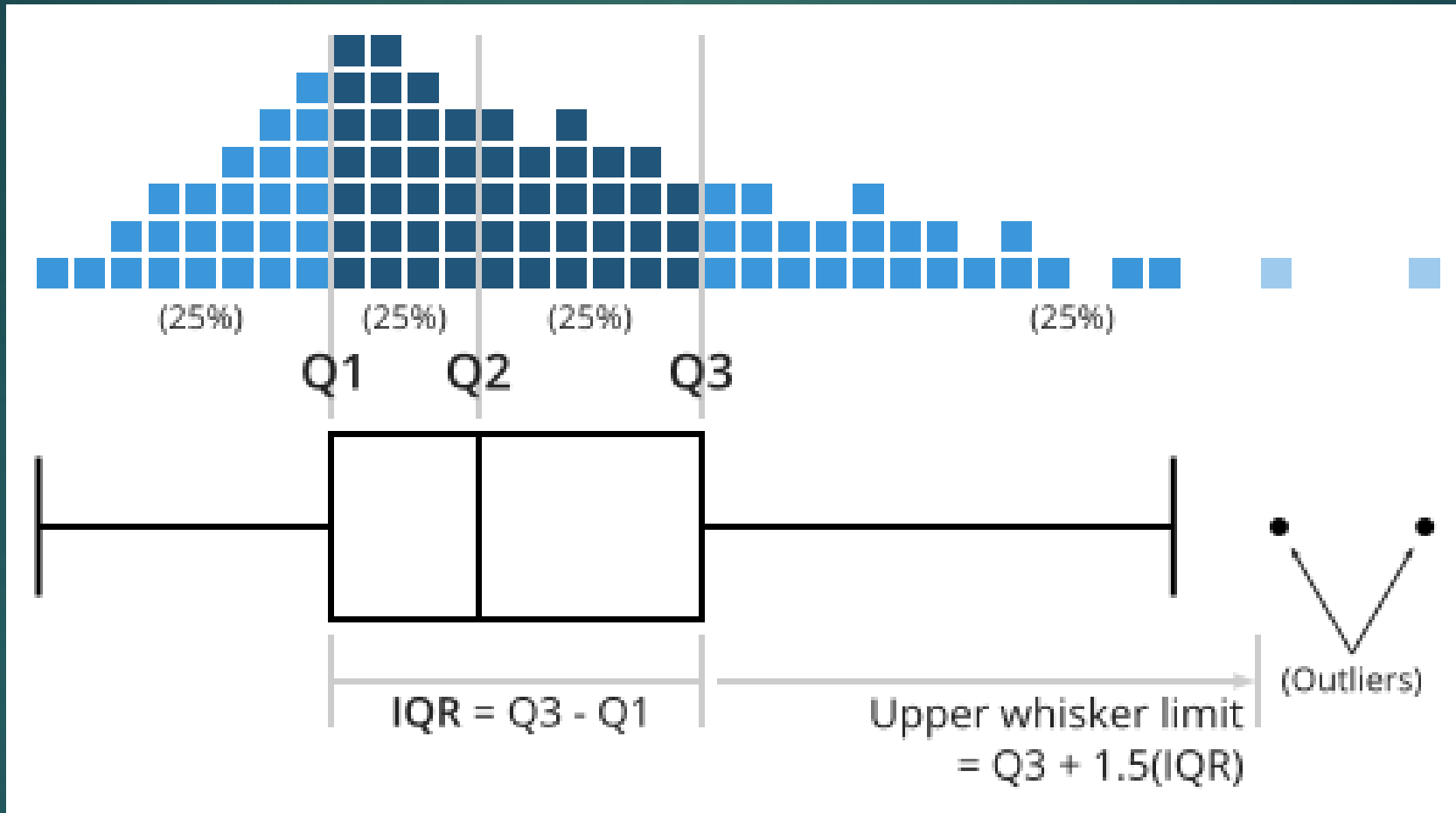
Between Q1 and Q3, is the interquartile range

In the boxplot, the solid line indicates the median and the dashed line indicates the mean.

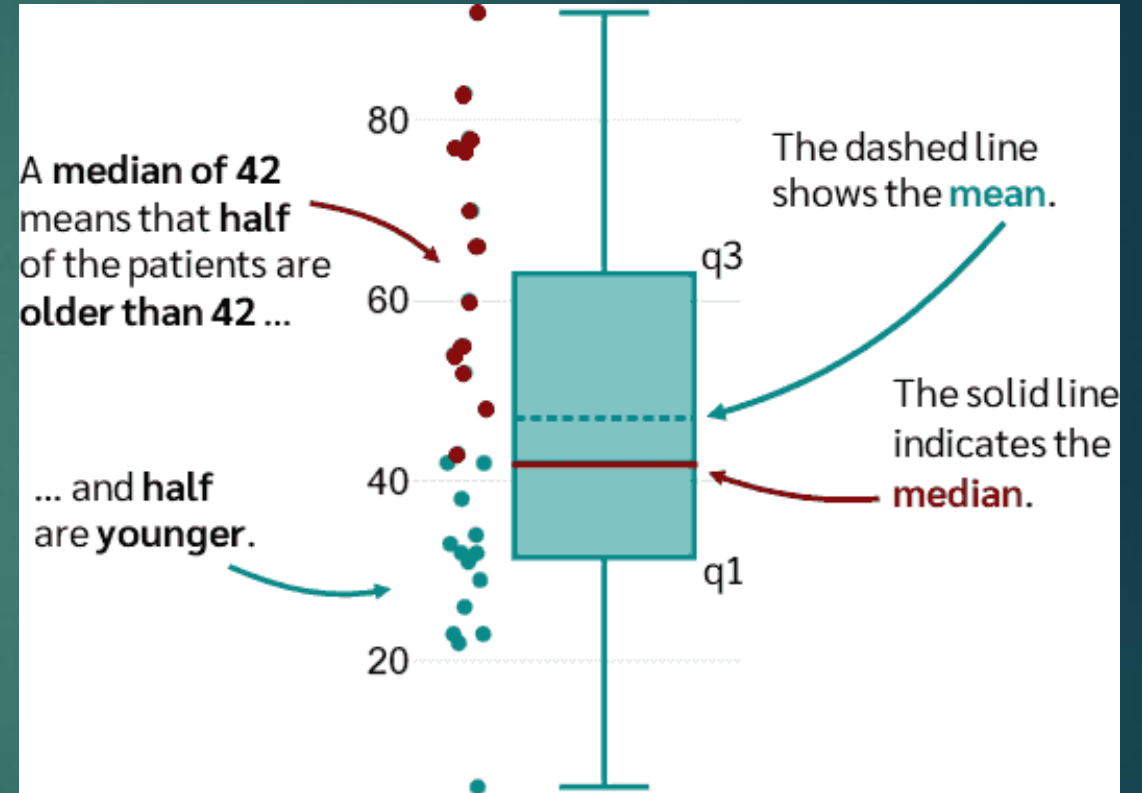
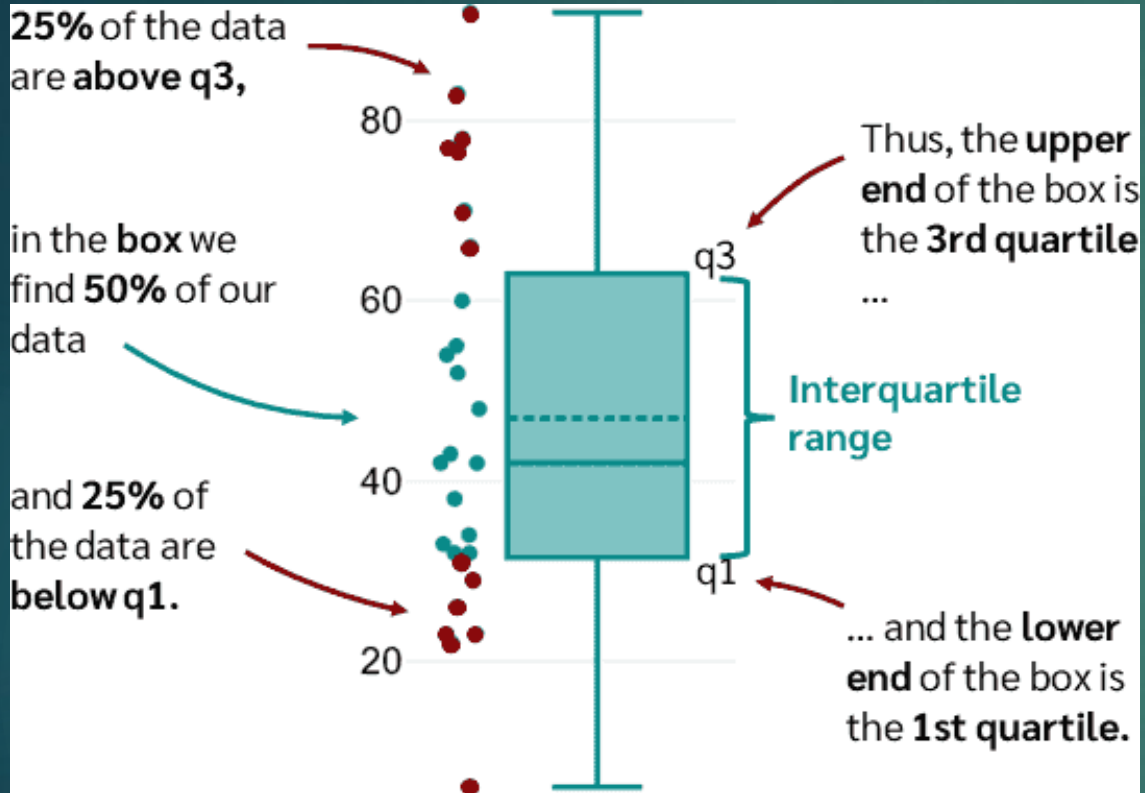
The T-shaped whiskers go to the last point, which is still within 1.5 times the interquartile range.

Points that are further away are considered extreme values (outliers).

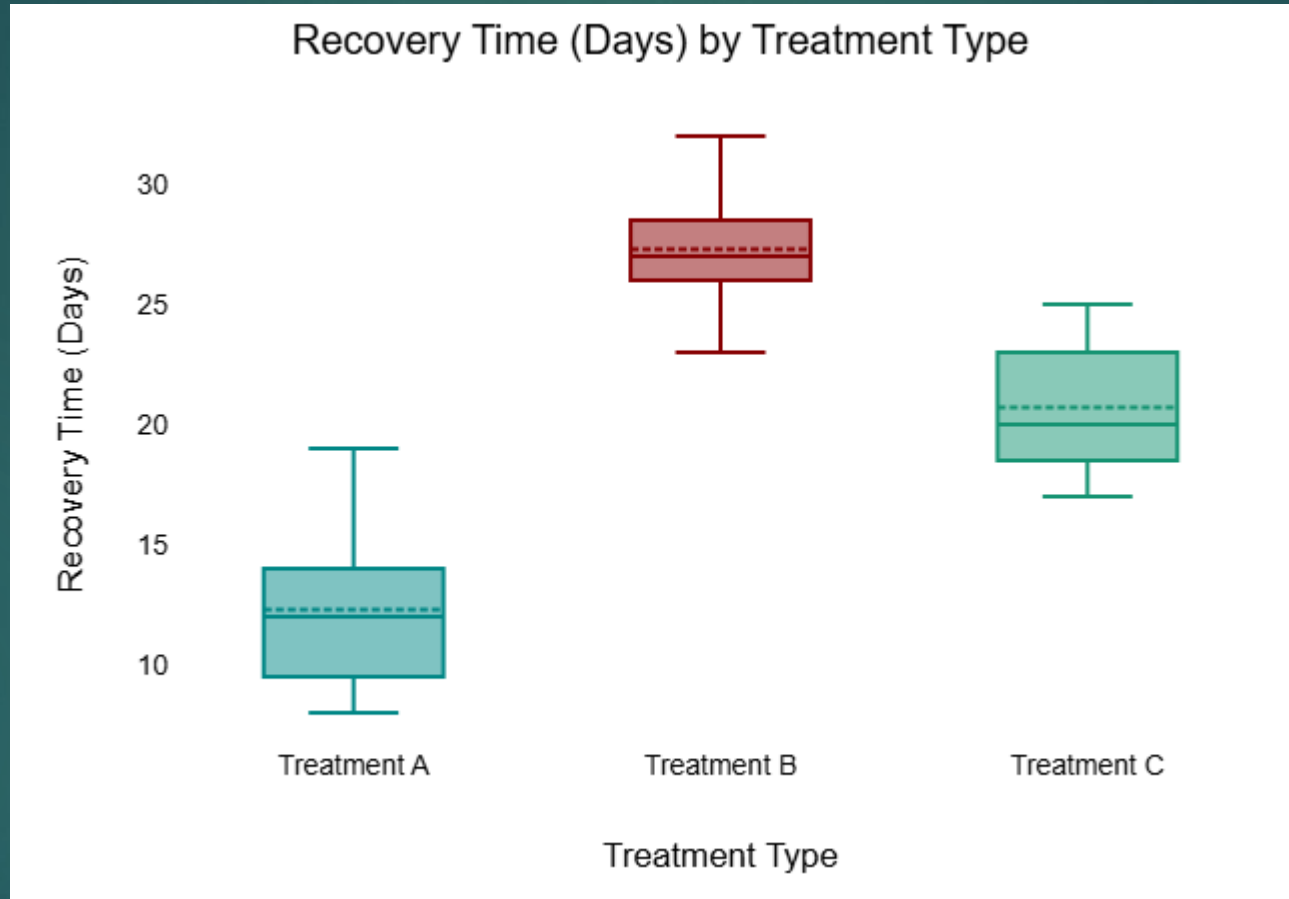
Boxplot



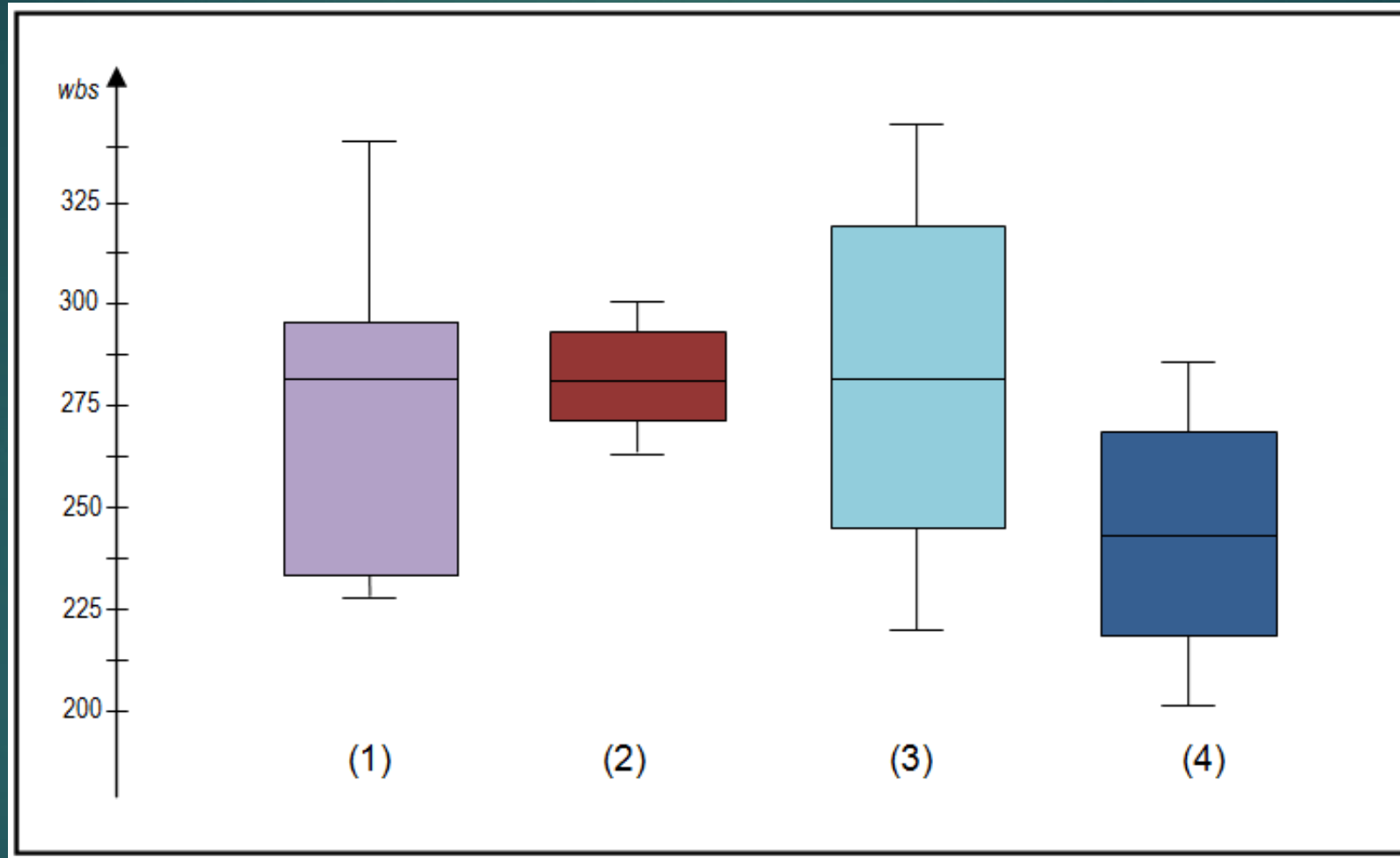
Boxplot



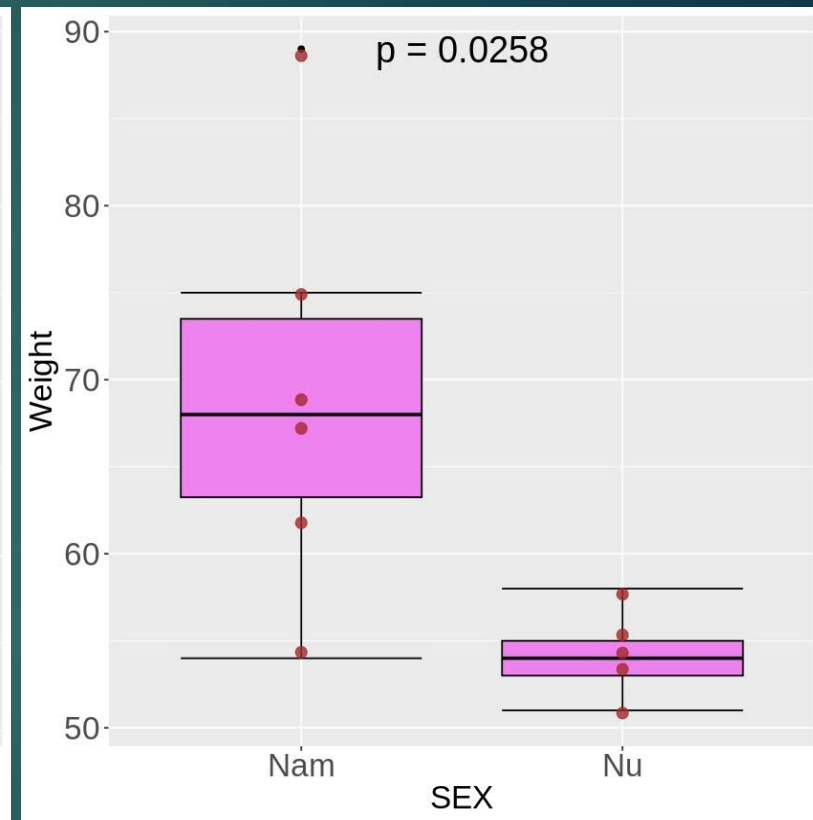
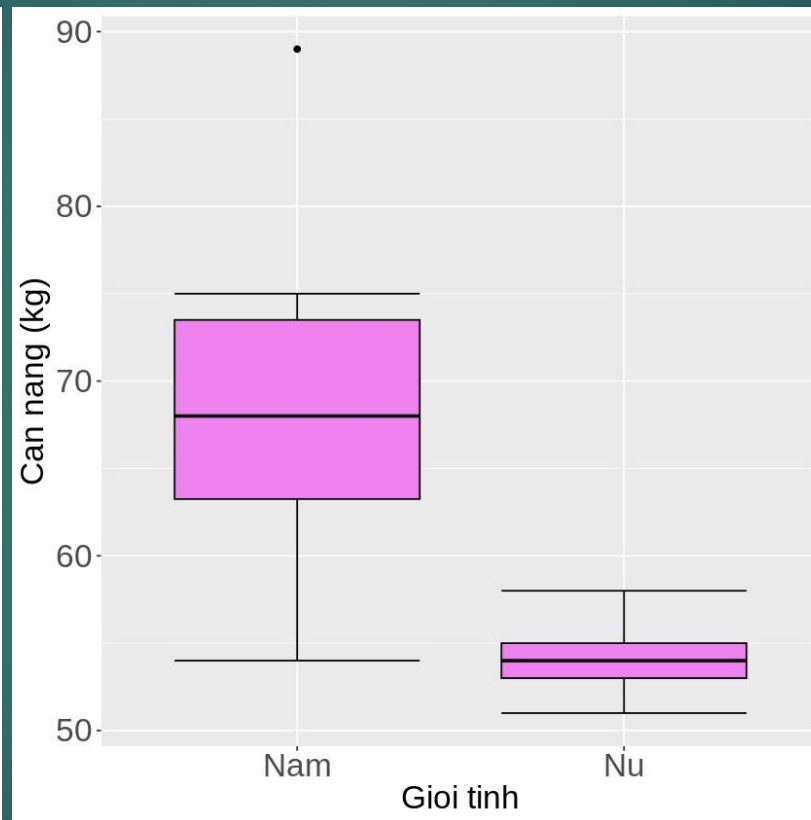
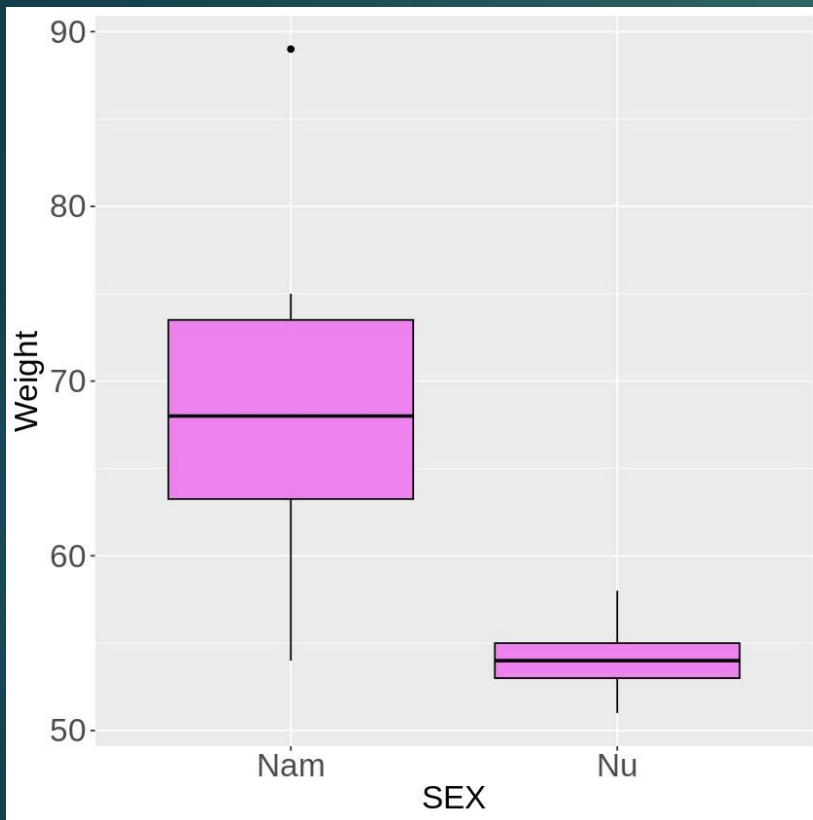
Boxplot



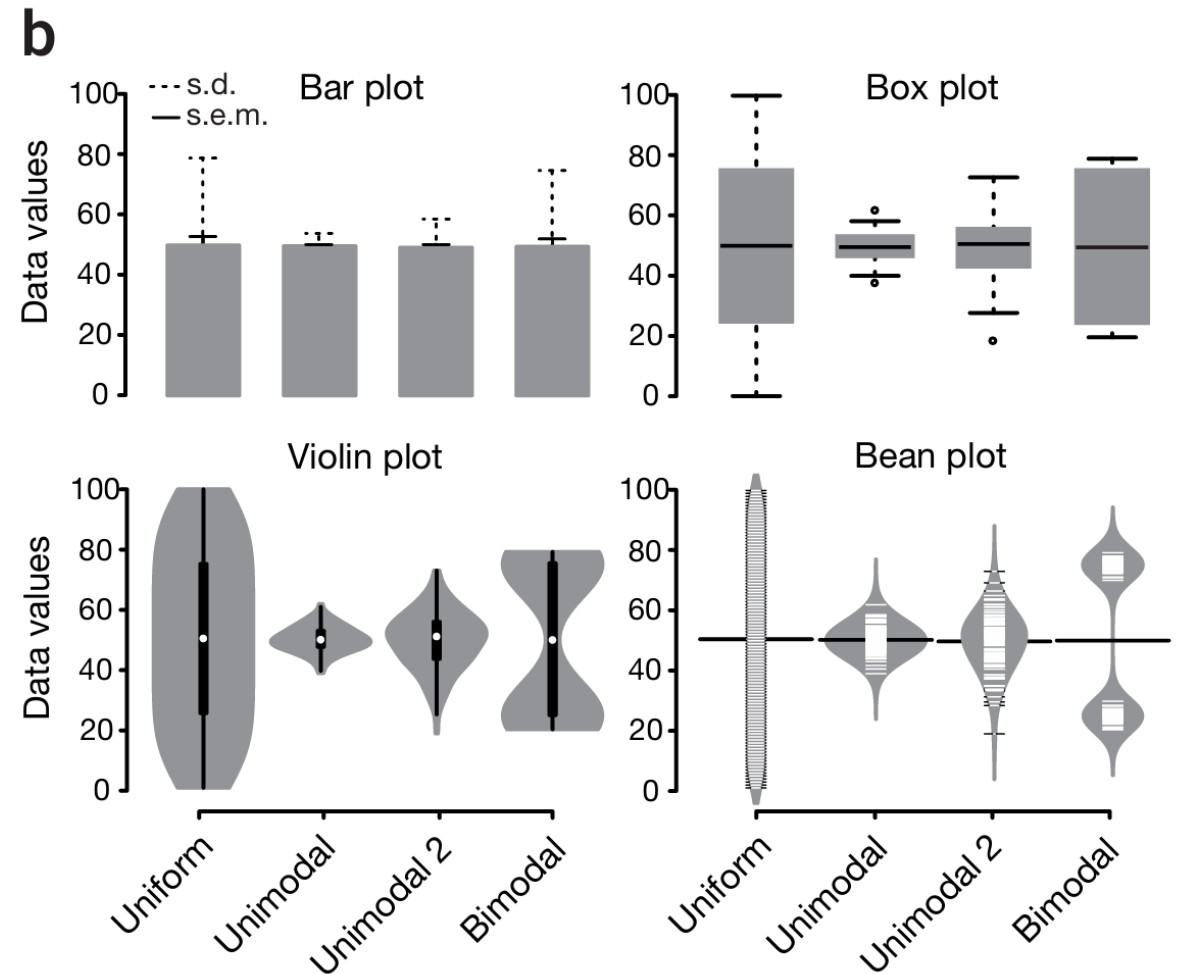
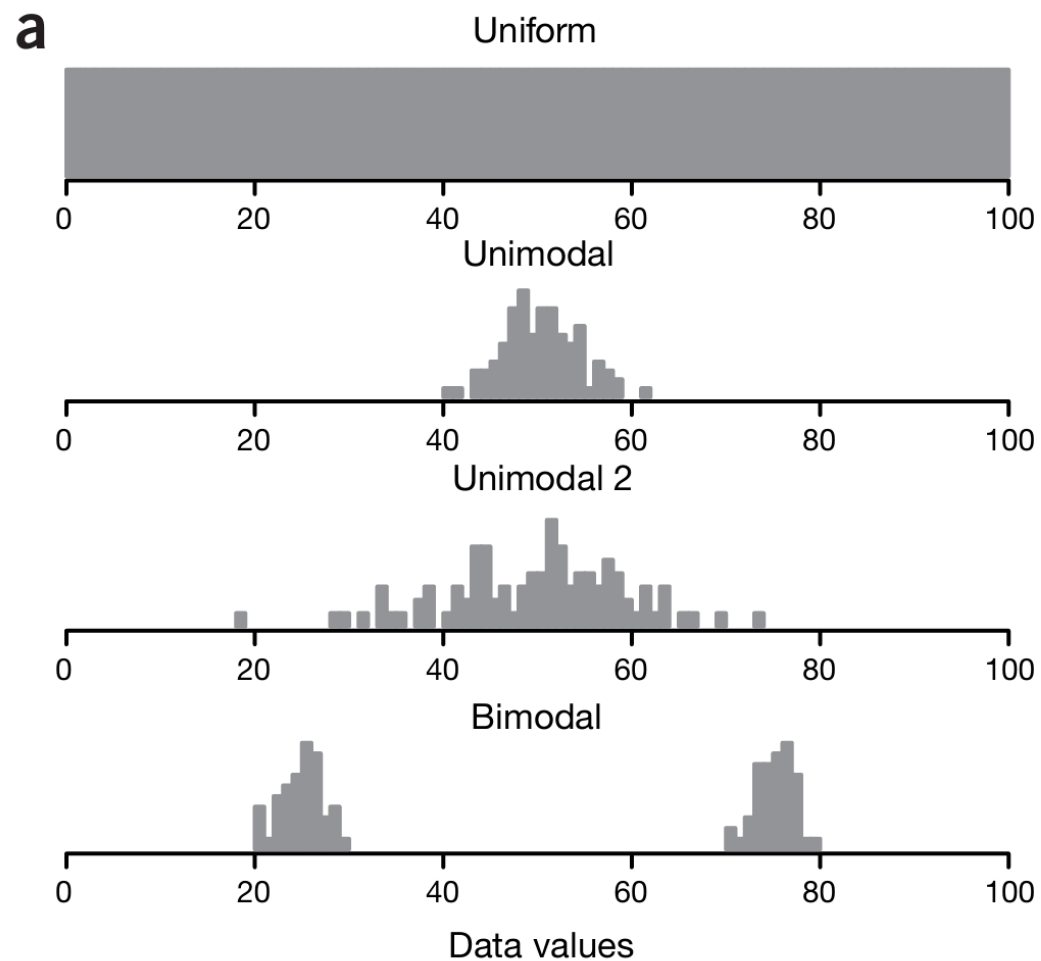
Boxplot



Boxplot



Boxplot and its Variants



Spitzer, M., Wildenhain, J., Rappsilber, J. *et al.* BoxPlotR: a web tool for generation of box plots. *Nat Methods* **11**, 121–122 (2014). <https://doi.org/10.1038/nmeth.2811>

Box Plot vs Violin Plot

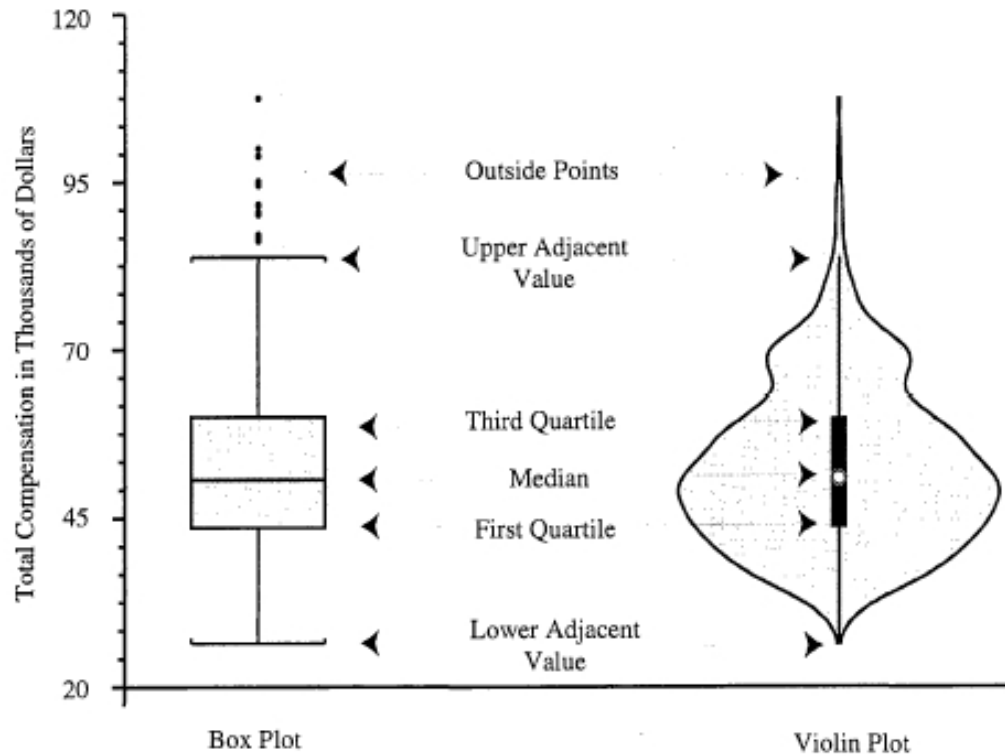
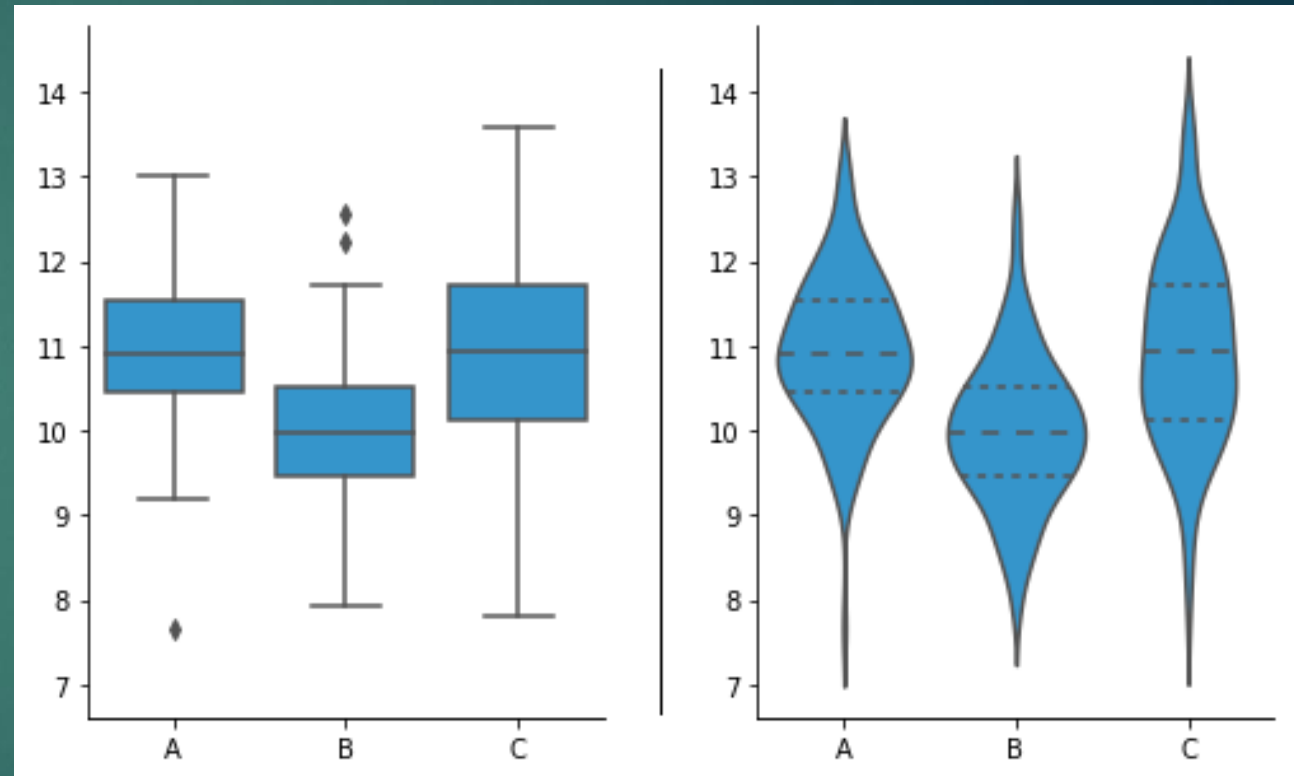
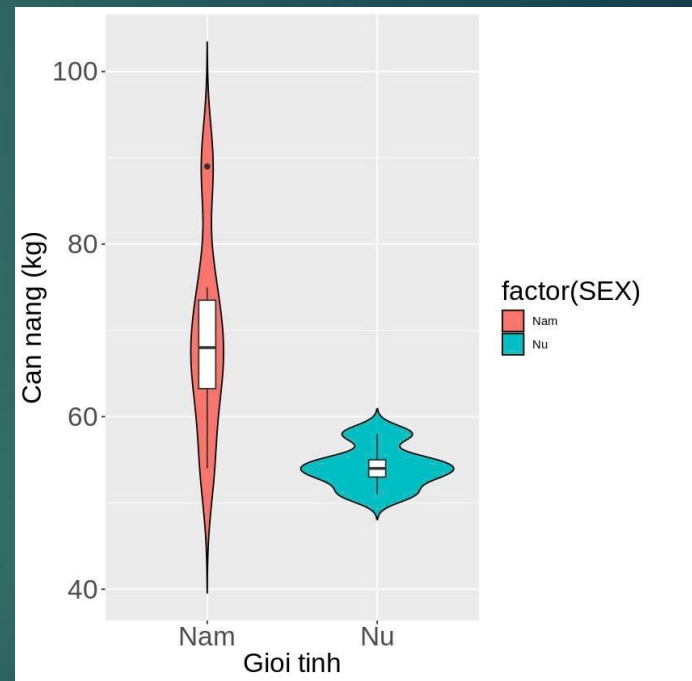
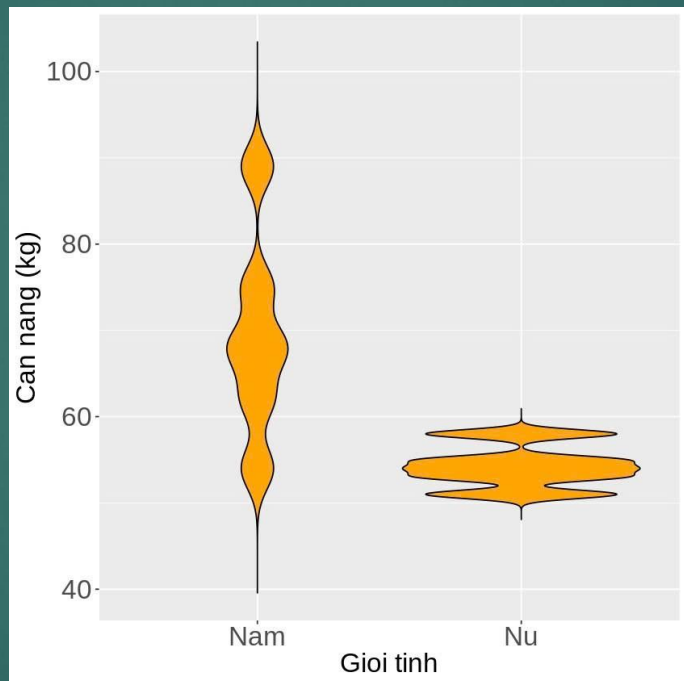
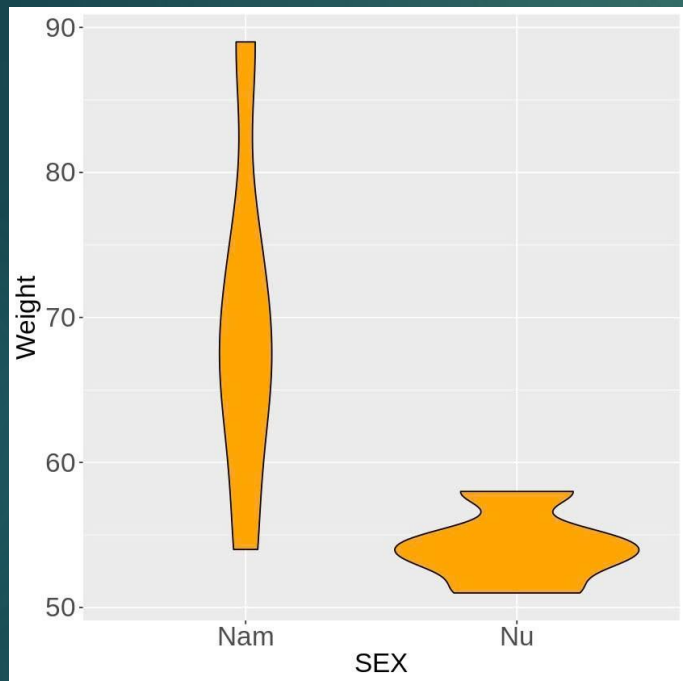


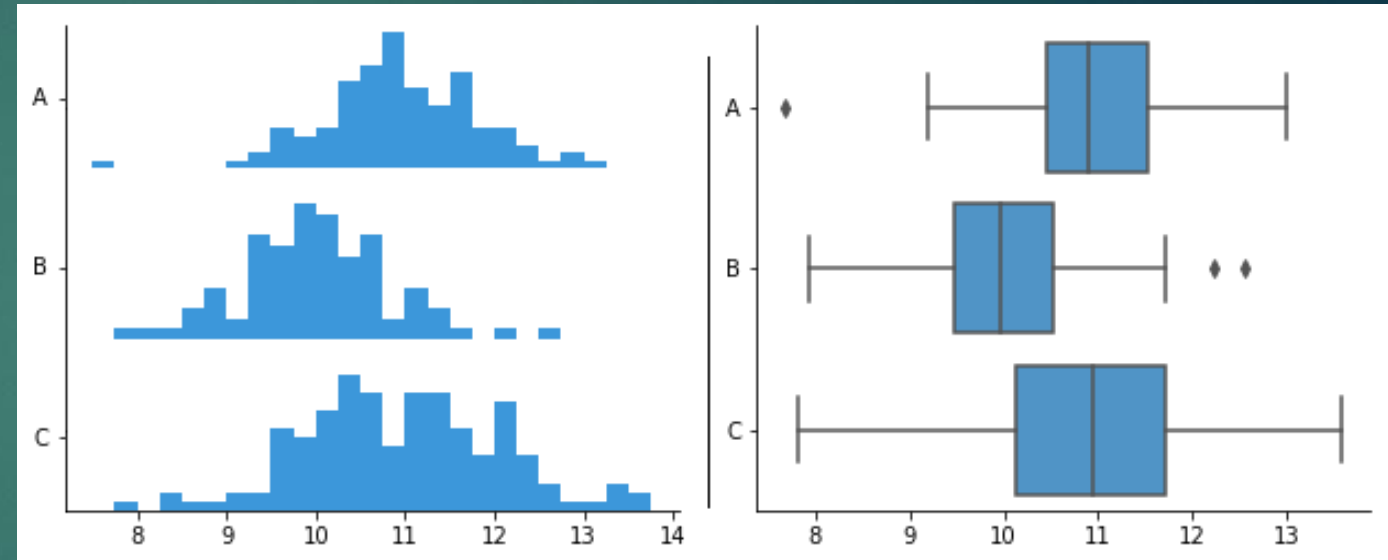
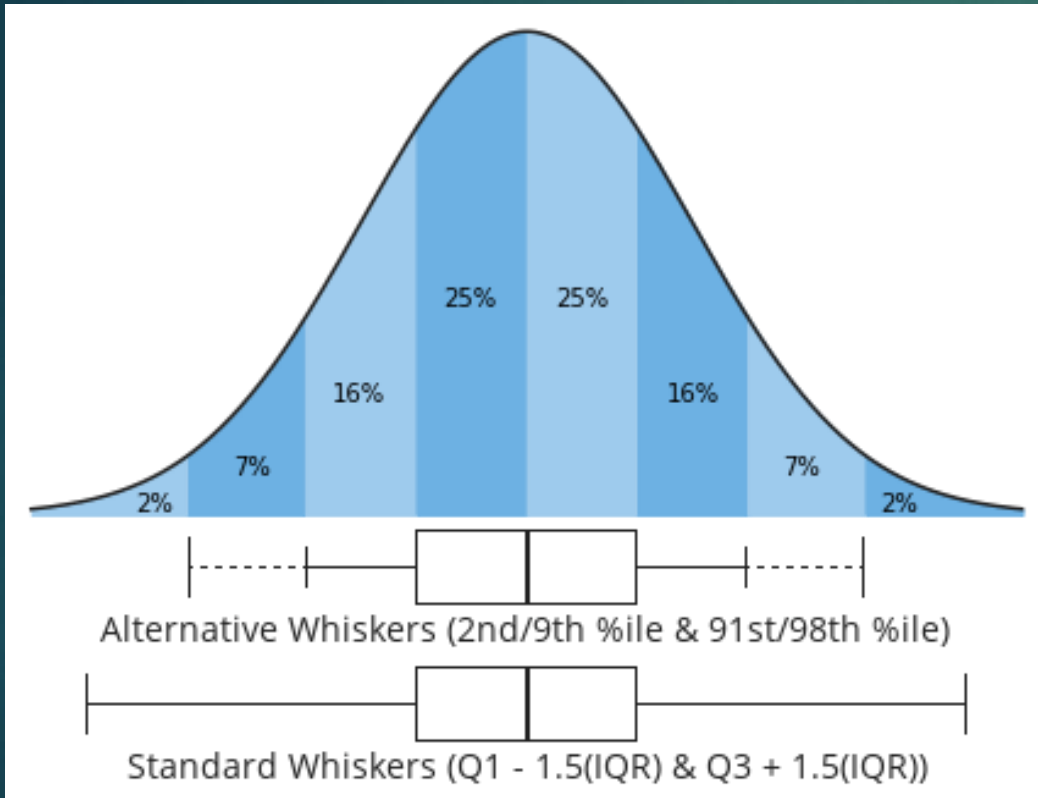
Figure 1. Common Components of Box Plot and Violin Plot. Total compensation for all academic ranks.



Violin Plot



Box Plot vs Histogram



Box Plot vs Bean Plot

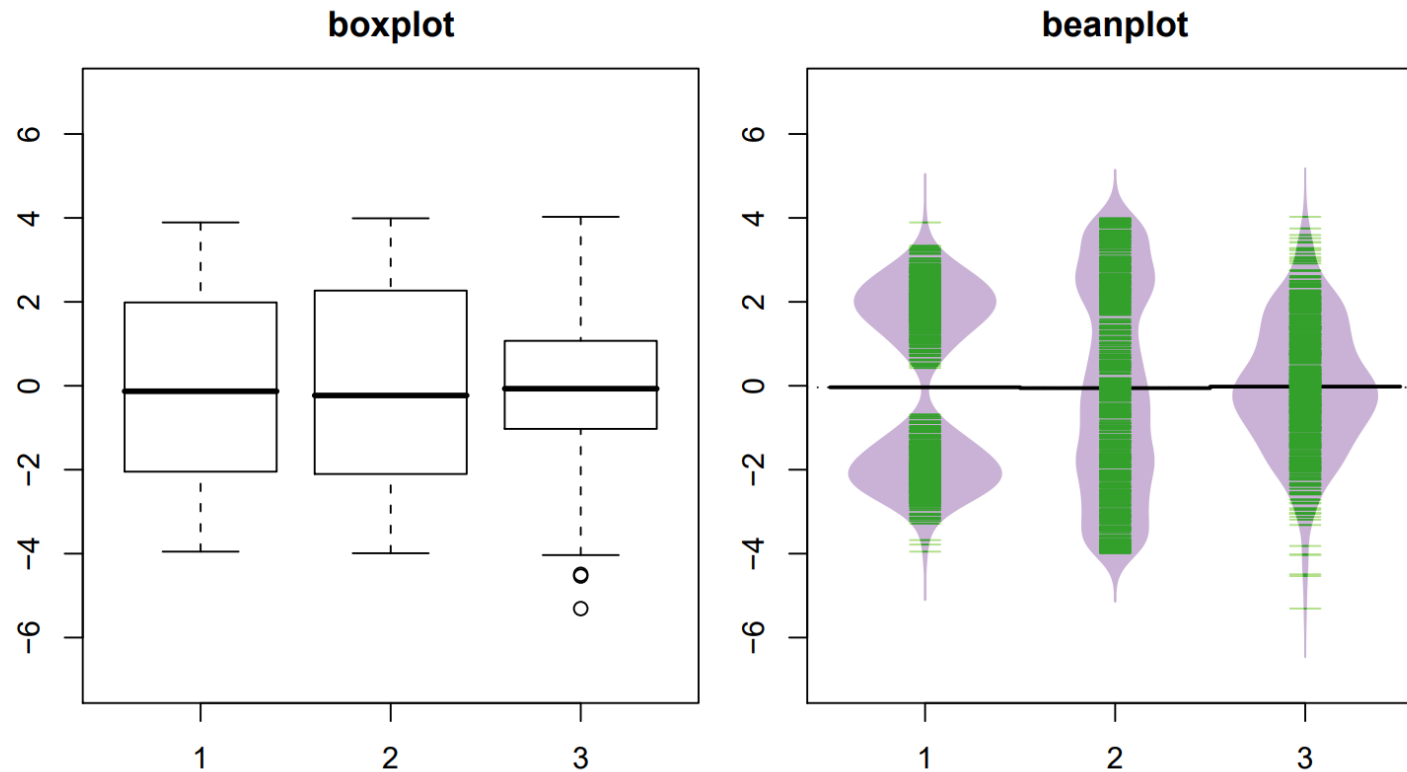


Figure 2: Plots for a bimodal, a uniform and a normal distribution. In the beanplot the green lines show individual observations, while the purple area shows the distribution.

Violin Plot vs Bean Plot

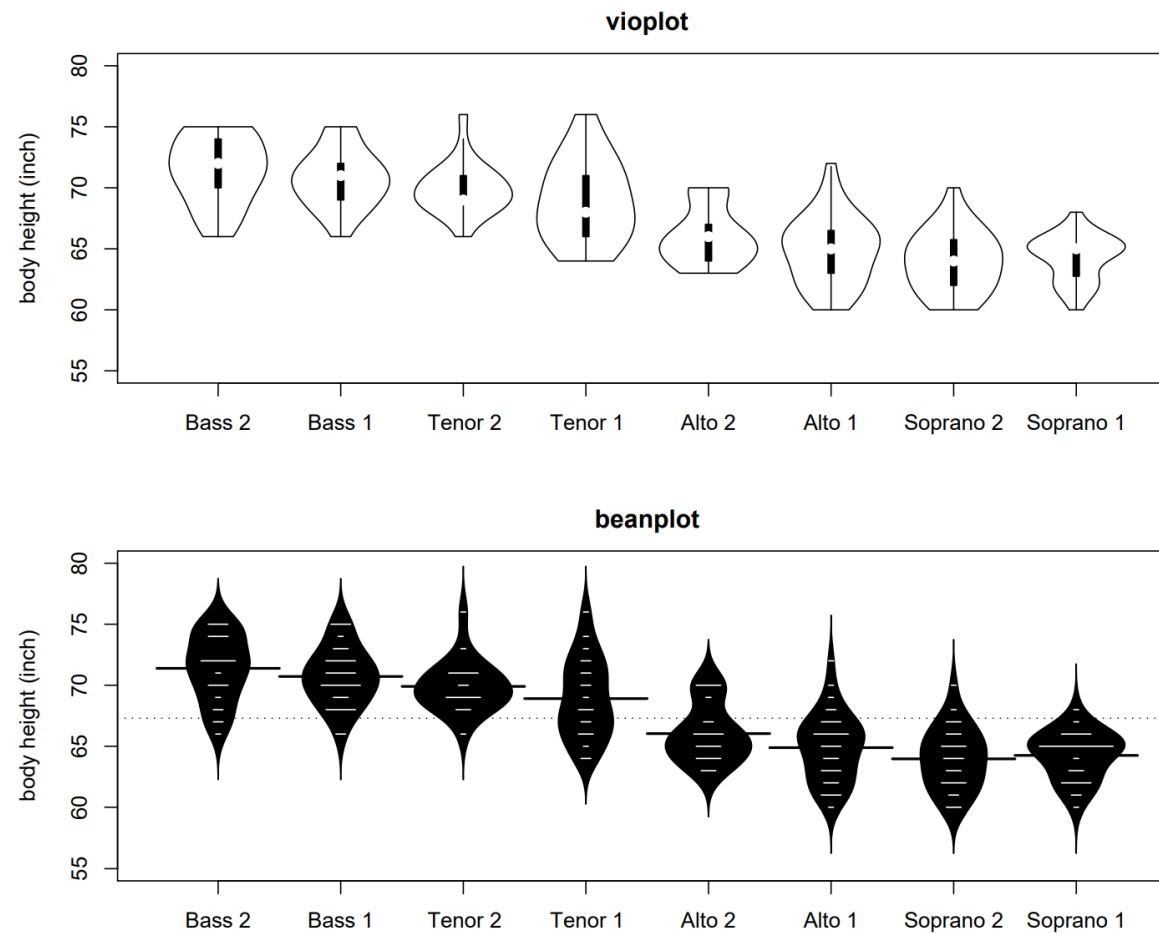


Figure 3: The height in inches of different singers. The beanplot clearly shows the individual measurements were rounded to whole inches.

Practice: Predict heart disease

Demographic:

- Sex: male or female(Nominal)
- Age: Age of the patient;(Continuous - Although the recorded ages have been truncated to whole numbers, the concept of age is continuous)

Behavioral

- Current Smoker: whether or not the patient is a current smoker (Nominal)
- Cigs Per Day: the number of cigarettes that the person smoked on average in one day.(can be considered continuous as one can have any number of cigarettes, even half a cigarette.)

Medical(history)

- BP Meds: whether or not the patient was on blood pressure medication (Nominal)
- Prevalent Stroke: whether or not the patient had previously had a stroke (Nominal)
- Prevalent Hyp: whether or not the patient was hypertensive (Nominal)
- Diabetes: whether or not the patient had diabetes (Nominal)

Medical(current)

- Tot Chol: total cholesterol level (Continuous)
- Sys BP: systolic blood pressure (Continuous)
- Dia BP: diastolic blood pressure (Continuous)
- BMI: Body Mass Index (Continuous)
- Heart Rate: heart rate (Continuous - In medical research, variables such as heart rate though in fact discrete, yet are considered continuous because of large number of possible values.)
- Glucose: glucose level (Continuous)

Predict variable (desired target)

- 10 year risk of coronary heart disease CHD (binary: "1", means "Yes", "0" means "No")



Thank you for your listening
Q & A